

MUHAMMAD MISHBAHUL HUDA

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Fresh graduate with a Diploma (D4) in Instrumentation Engineering from Institut Teknologi Sepuluh Nopember (ITS), GPA 3.56/4.00. Hands-on experience in instrument maintenance, calibration, and troubleshooting gained through an internship in the Instrument Maintenance II section at PT Petrokimia Gresik. Strong interest in mechanical and electrical systems, DIY fabrication, and design, with proven ability to work effectively in field environments and cross-functional teams. Certified in Engineer scheme, low-voltage electrical installation and PLC-based electronic equipment operation.

EDUCATIONAL LEVEL

Institut Teknologi Sepuluh Nopember (ITS) — Surabaya, Indonesia
*Bachelor of Instrumentation Engineering, **GPA**: 3.56/4.00* — *July 2022 - Aug 2026*

- Thesis: Integrated Digital Wastewater Flow Monitoring System to Support Sustainable Water Management at ITS.
- Relevant coursework: Measurement & Calibration Technology, Automation Engineering, DCS & SCADA, Predictive Maintenance, Troubleshooting, Rotating Equipment, Instrumentation Systems Interconnection, Engineering Commissioning.

SMK Negeri 1 Cerme — Gresik, Indonesia
Vocational High School of Industrial Automation Engineering — *July 2018 - June 2022*

- Gained hands-on experience through laboratory projects, heavy equipment internship, and community activities.

PROFESSIONAL EXPERIENCE

PT Petrokimia Gresik — Gresik, Indonesia
Instrument Maintenance Intern — *Feb 2026 - Jul 2026*

- Performed instrument calibration, troubleshooting, and maintenance activities.
- Developed a fertilizer moisture monitoring device for NPK products to support compliance with SNI 2803, achieving 98.13% accuracy compared with NIR Proximate (Büchi Labortechnik AG) as reference instrument.

Seismic Resilience Laboratory, Dept. of Civil and Construction, NTUST — Taipei City, Taiwan
Research Assistant — *Aug 2024 – Sept 2024*

- Selected for Taiwan's International Internship Pilot Program (IIPP) and conducted testing of an earthquake detection system with multi-hop sensor network.

Electronic Laboratory, Dept. Instrumentation Engineering, ITS — Surabaya, Indonesia
Student Coordinator - Mechanical Manager - Mechanical Staff — *Nov 2023 – Jan 2026*

- Supervised 14 laboratory assistants, 4 Instrumentation courses, supporting practicum activities for 200+ students and coordinating research project with external partners under INSPIRASI, HETI, and Kedaireka.
- Designed and developed 3D mechanical components and assemblies for instrumentation projects, including complete device structures, sensor brackets, and controller enclosures using Autodesk Inventor, while performing maintenance of 3D printers.

PT Mentari Lima Teknik — Surabaya, Indonesia
Helper Technician Intern — *Nov 2019 – Apr 2020*

- Performed hoist and crane maintenance by replacing electromechanical components, elevator inspection, and assisting girder fabrication and hoist installation.

PROJECT EXPERIENCE

Integrated Digital Wastewater Flow Monitoring System to Support Sustainable Water Management at ITS Smart Eco Campus

- Designed and Developed an integrated flow monitoring using a Bürkert S030 flow meter, and ControlByWeb X-404 with Modbus RTU Communication, achieving 95.46% measurement accuracy through validation against TUF-2000B ultrasonic flow meter.
- Presented the research at International Conference on Environmental Engineering (ICOEE) 2026.

Soil Moisture Monitoring System for Fertilizer product at NPK II, III, & IV Bagging Area PT Petrokimia Gresik

- Designed and implemented a NPK fertilizer moisture monitoring system using MT-02A Soil Moisture Sensor, Seeeduino XIAO, and Modbus RTU Communication for NPK bagging area. Validated the system a NIR Proximate method at Factory Laboratory.

Monitoring System of Dissolved Oxygen Concentration at Oxidation Ditch IPLT Keputih Surabaya

- Fabricated panel, installed electrical wiring including DO (Atlas Scientific), Ammonia NHN-046 and Nitrate NON-406 (Chemins)
- Designed and 3D-printed custom sensor mounting brackets, ensuring proper positioning and mechanical integration.

CERTIFICATION

- **LSP P1 BNSP** – Engineer Scheme Competency (LSP ITS)
- **LSP P3 BNSP** – PLC-Based Electronic Equipment Operation
- **LSK** – Low Voltage Electrical Installation Competency (PT Lisan Nusantara Satu)

ORGANIZATIONAL EXPERIENCE

- Journalistic Coordinator (2020-2022), Facilitator (2022-now) of Komunitas Anak Gresik, KBP3A Gresik
- Volunteer of Iling Bumi Gresik (2022-now)
- Instructure Committee of Cadres Element, HIMATEKINS ITS (2024-2025)
- Secretarial Staff of GERIGI ITS 2023
- Logistic Staff of VONDASI 5.0 ITS 2023, Faculty of Vocational Studies
- Decoration Staff of ITS Expo 2023 and Maba Cup 2022
- Income Staff (2023), Instraining Coordinator (2024) of Instrumentation Festival, Dept. of Instrumentation Engineering

SKILL

Technical Skills

- CAD & Visual Design: AutoCAD, Autodesk Inventor; Visual design with CorelDRAW, Figma
- 3D Printing: Creality Ender 2/3, CR-10 Max, ELEGOO Neptune 4 Plus
- Industrial Communication: Modbus TCP/IP, Modbus RTU
- Programming: Arduino IDE, Visual Studio Code (Python), JetBrains IDE (Rust), Eclipse (C++), ModusToolbox (PSoC 6)
- Engineering Software: MATLAB, Ansys, Aspen HYSYS, Festo FluidSIM
- PLC Programming: Omron CX- One, Siemens TIA Portal, Mitsubishi GX Works
- Office & Productivity: Microsoft Word, Excel, PowerPoint

Soft Skills

- Teamwork
- Communicative
- Adaptive
- Problem Solving
- Leadership

Languages

- Indonesian (Native)
- English (Intermediate)